

Antenna Design Steps

Step 1: Patch Width (W)

The width of the rectangular patch is given by:

$$W = \frac{c}{2f_o \sqrt{\frac{\epsilon_r + 1}{2}}} \quad (1)$$

where c represents the velocity of light in free space.

Step 2: The Effective Dielectric Constant (ϵ_{eff})

The ϵ_{eff} is computed as:

$$\epsilon_{eff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left(1 + 12 \frac{h}{w} \right)^{-1/2} \quad (2)$$

This accounts for the fringing fields at the patch periphery.

Step 3: Effective Length (L_{eff})

The patch effective length is given by:

$$L_{eff} = \frac{c}{2f_o \sqrt{\epsilon_{eff}}} \quad (3)$$

Step 4: Length Extension (ΔL)

$$\Delta L = 0.412h \frac{(\epsilon_{eff} + 0.3) \left(\frac{w}{h} + 0.264 \right)}{(\epsilon_{eff} - 0.258) \left(\frac{w}{h} + 0.8 \right)} \quad (4)$$

Step 5: Length of Patch (L)

The actual patch length is obtained as:

$$L = L_{eff} - 2\Delta L \quad (5)$$

Step 6: Substrate Dimensions (Ls & Ws)

$$L_s = L + 2 * \Delta L \quad (6)$$

$$W_s = W + 2 * \Delta L \quad (7)$$